CS 540 Introduction to Artificial Intelligence

Classification - KNN and Naive Bayes

University of Wisconsin-Madison

Spring 2025

Announcements

- **Homework:**
  - HW5 online, due Monday March 3rd at 11:59 PM
  - Linear Regression

- Class roadmap:

Machine Learning: kNN& Naive Bayes
Machine Learning: Neural Networks I (Perceptron)
Machine Learning: Neural Networks II
Machine Learning: Neural Networks III

Supervised Learning

Outline

- K-Nearest Neighbors
- Maximum likelihood estimation
- Naive Bayes

Part I: K-nearest neighbors

☆ (KNN ≠ K-Means)

classification Technique

clustering Technique

WIKIPEDIA
The Free Encyclopedia

Main page

Article Talk

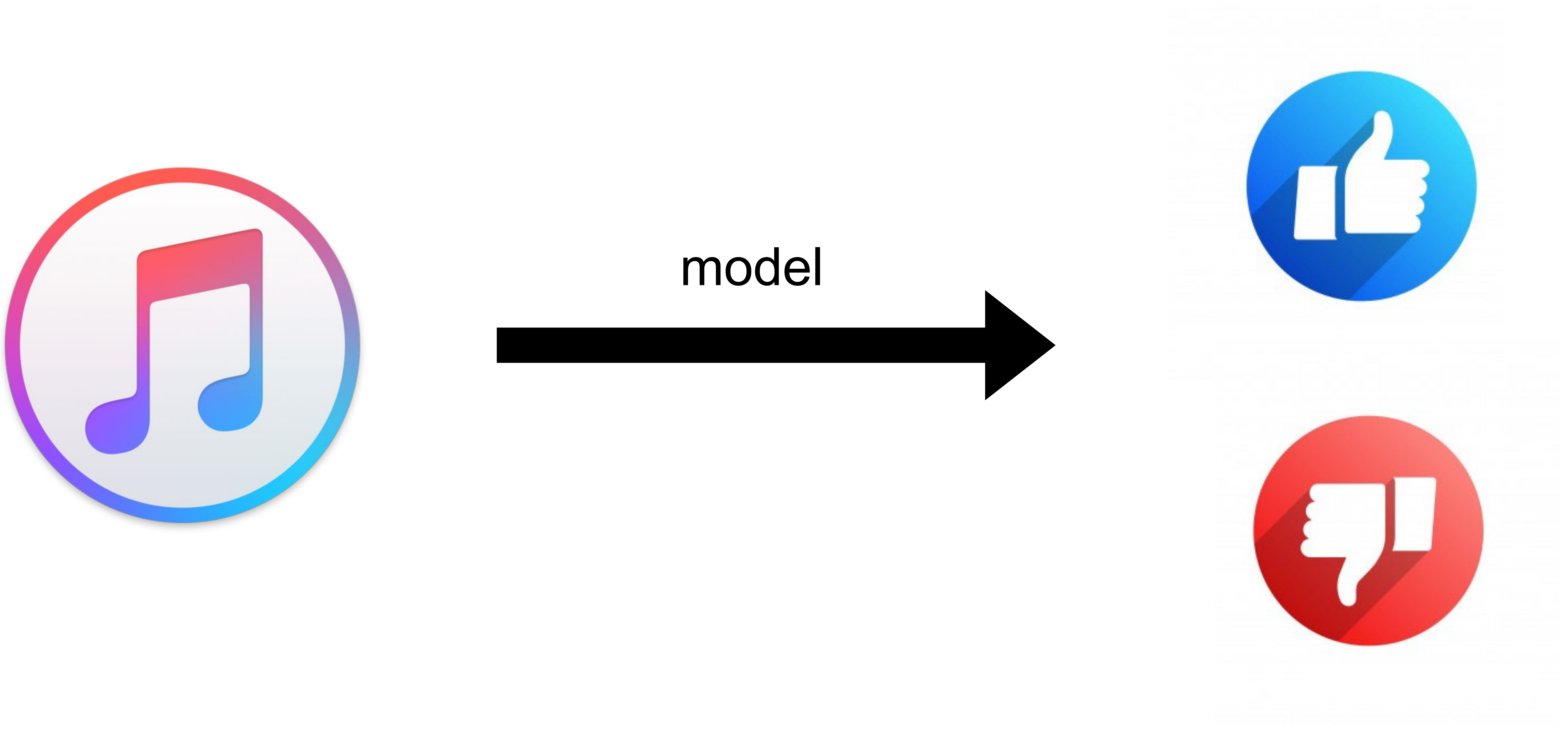
k-nearest neighbors algorithm

From Wikipedia, the free encyclopedia

Not to be confused with k-means clustering.

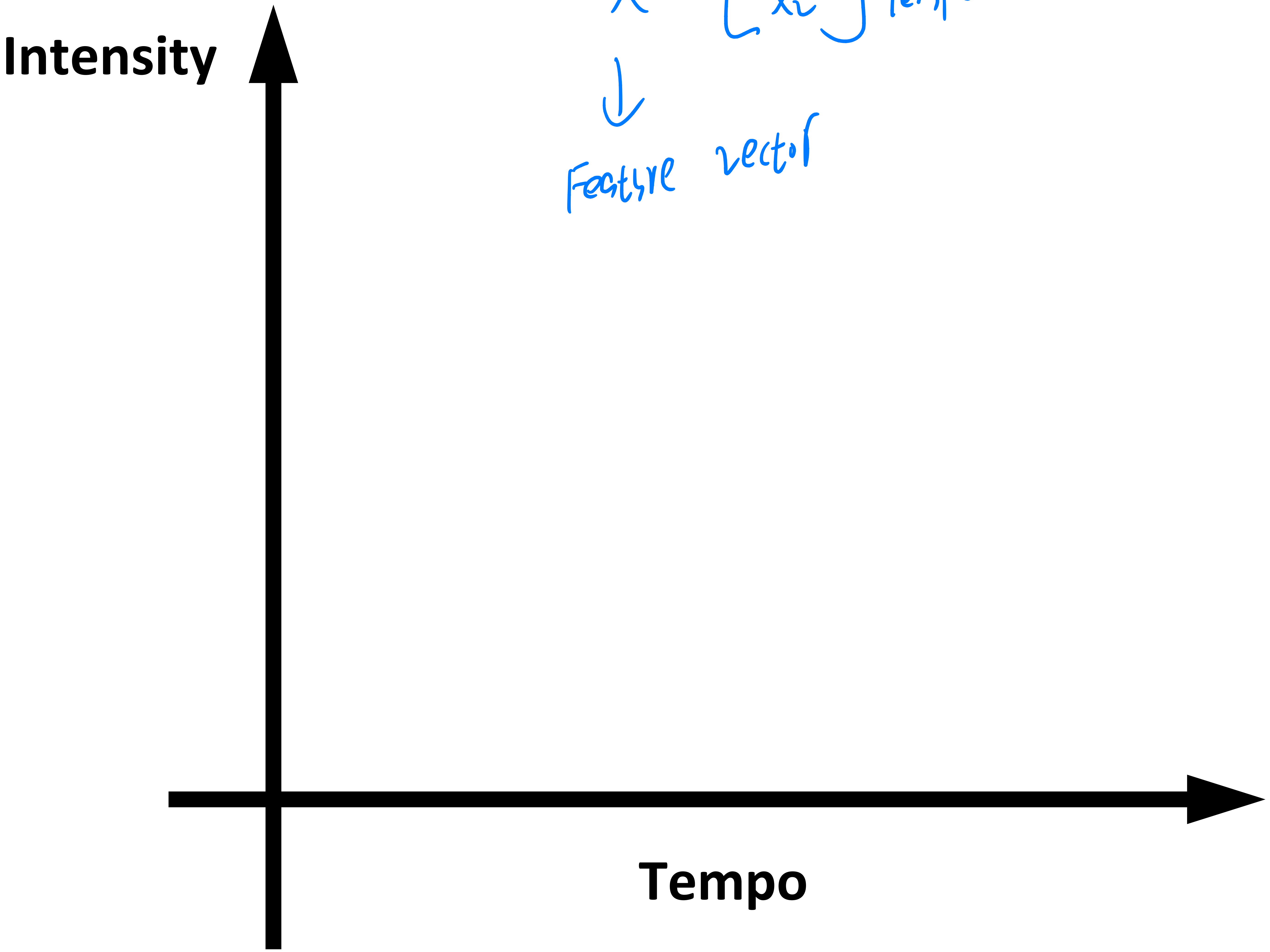
(source: wiki)

Example 1: Predict if a user likes a song or not



Example 1: Predict if a user likes a song or not

User Sharon



$$X = \begin{array}{c} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \end{array} \begin{array}{l} \text{Intensity} \\ \text{Tempo} \end{array}$$

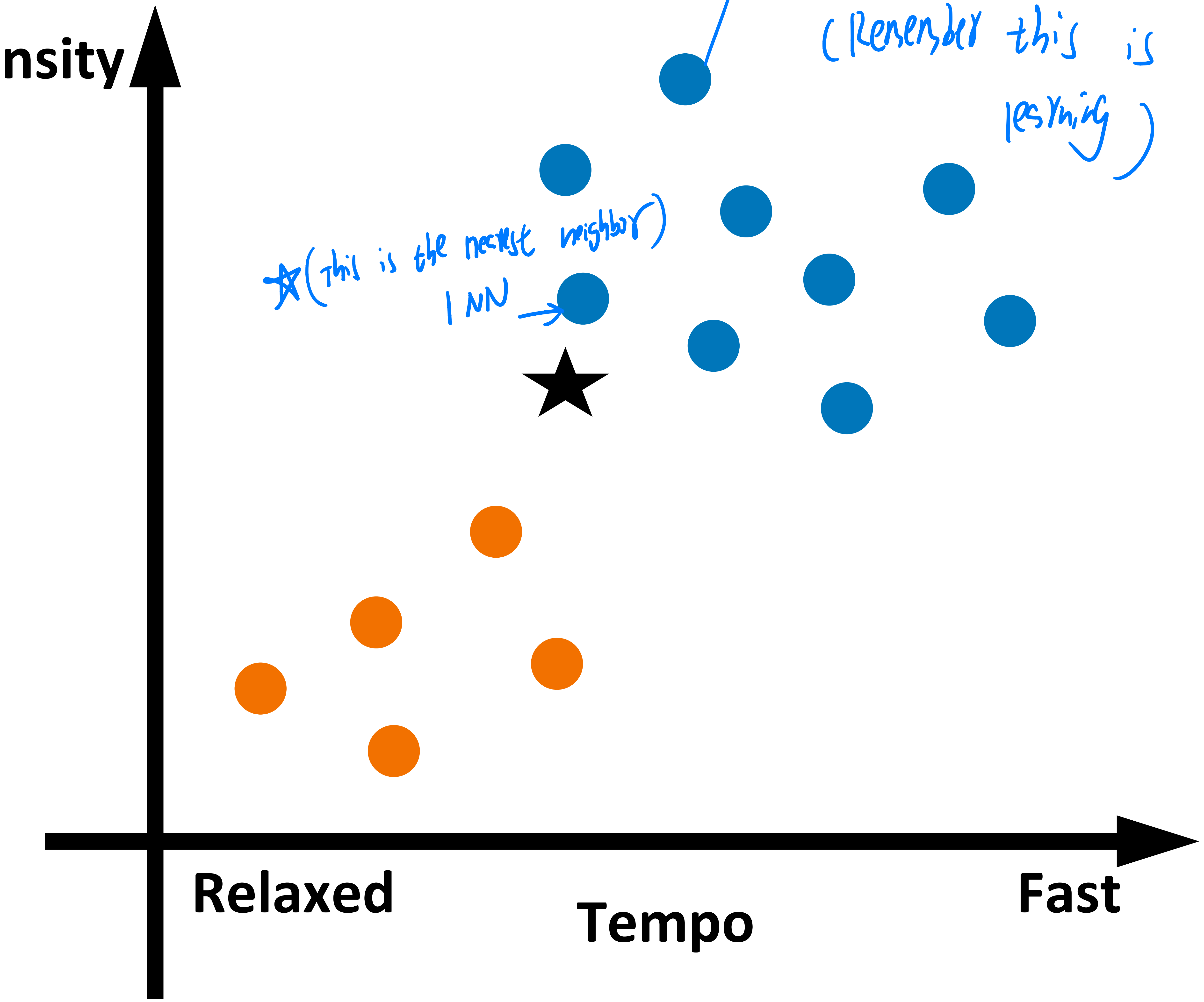
$\downarrow$

Feature vector

★ 1 nearest neighbor in this case

Example 1: Predict if a user likes a song or not 1-NN

★ The data themselves are with labels
(Remember this is a supervised learning)

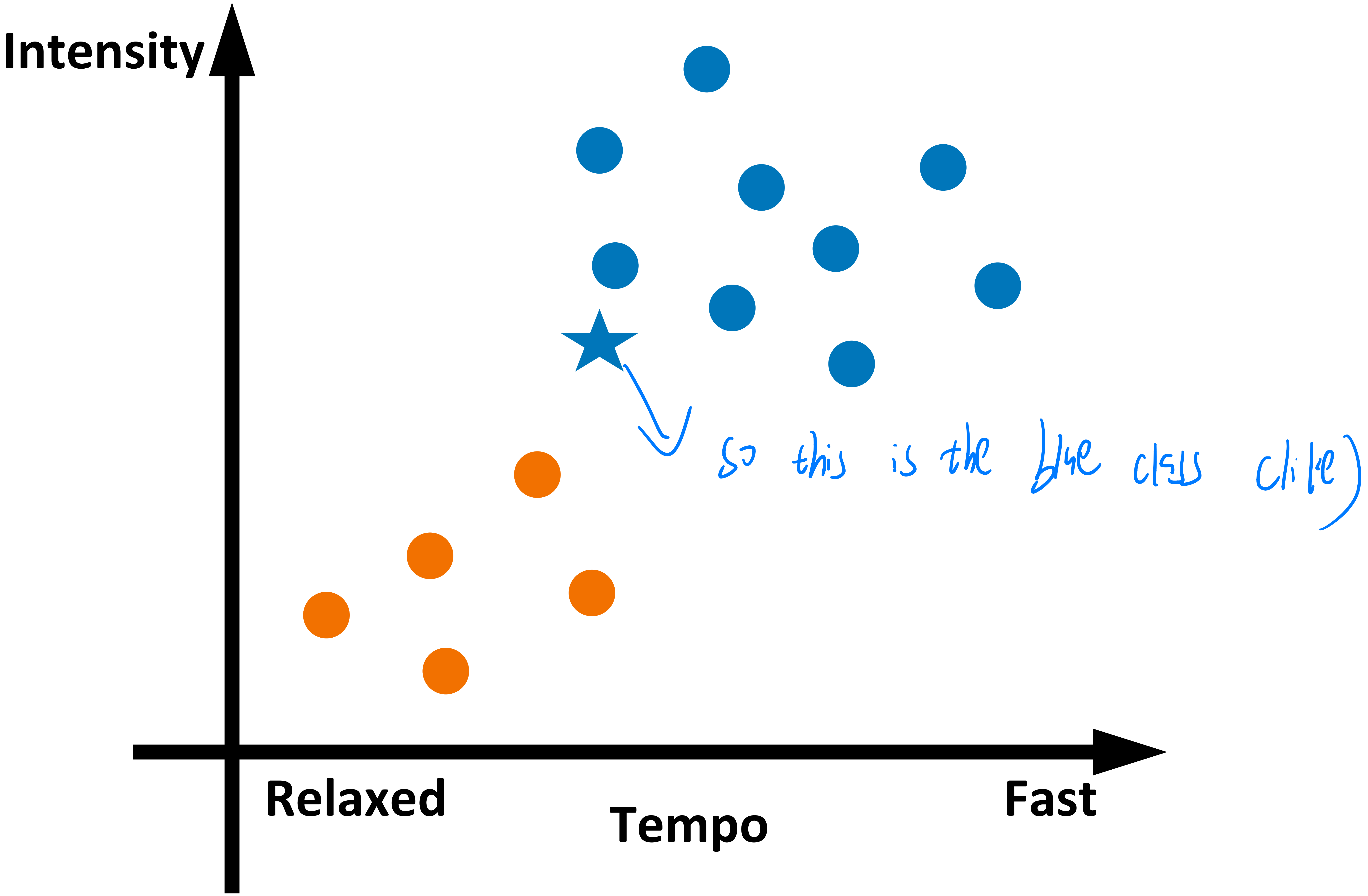


Example 1: Predict if a user likes a song or not 1-NN

User Sharon

● Dislike

● Like



K-nearest neighbors for classification

Label

- **Input**: **Training data** $(\mathbf{x}_1, y_1), (\mathbf{x}_2, y_2), \ldots, (\mathbf{x}_n, y_n)$

  Distance function $d(\mathbf{x}_i, \mathbf{x}_j)$; **number of neighbors** k; **test data** $\mathbf{x}^*$

1. Find the k training instances $\mathbf{x}_{i_1}, \ldots, \mathbf{x}_{i_k}$ closest to $\mathbf{x}^*$ under $d(\mathbf{x}_i, \mathbf{x}_j)$
2. Output y^*, the majority class of $y_{i_1}, \ldots, y_{i_k}$. Break ties randomly.

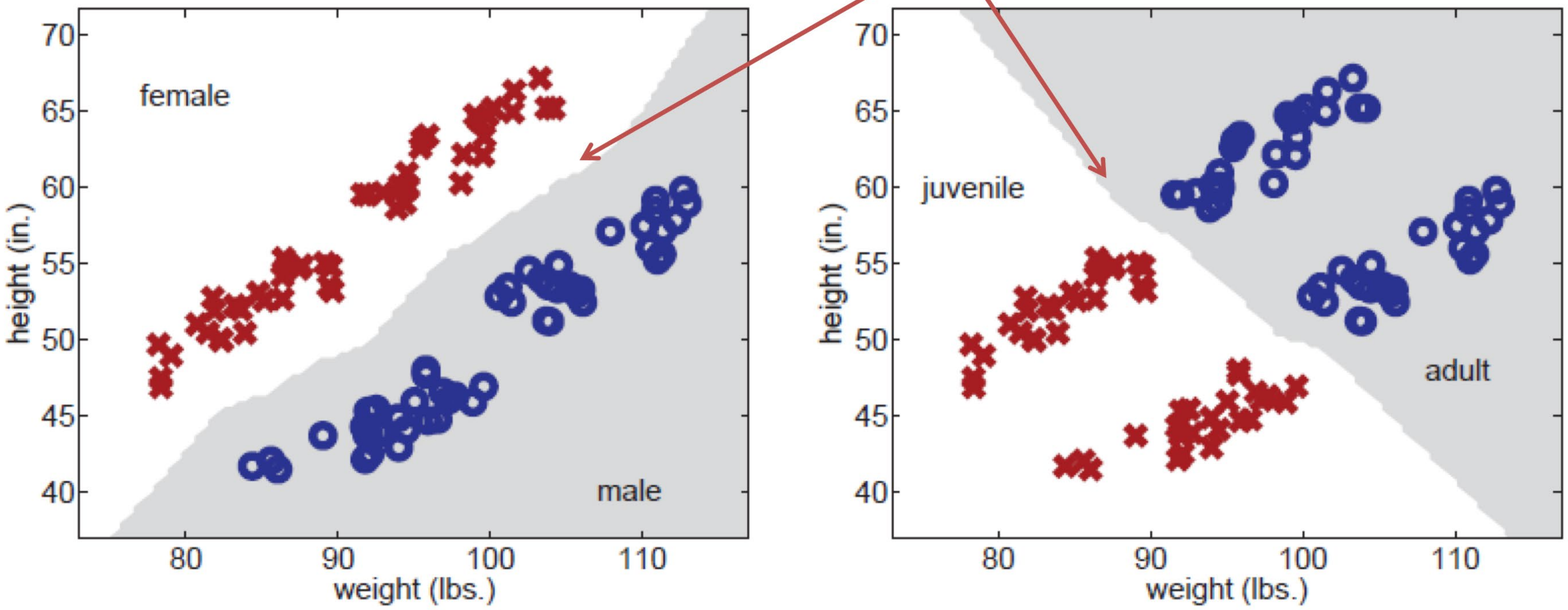
☆ The only output

(☆ Among these k nearest neighbors ($x_{i_1} \ldots x_{i_k}$), check their corresponding $y_{i_1} \ldots y_{i_k}$ and find the most frequent one, assign y^* to it)

Example 2: 1-NN for little green man

- Predict gender (M,F) from weight, height
- Predict age (adult, juvenile) from weight, height

Decision boundary



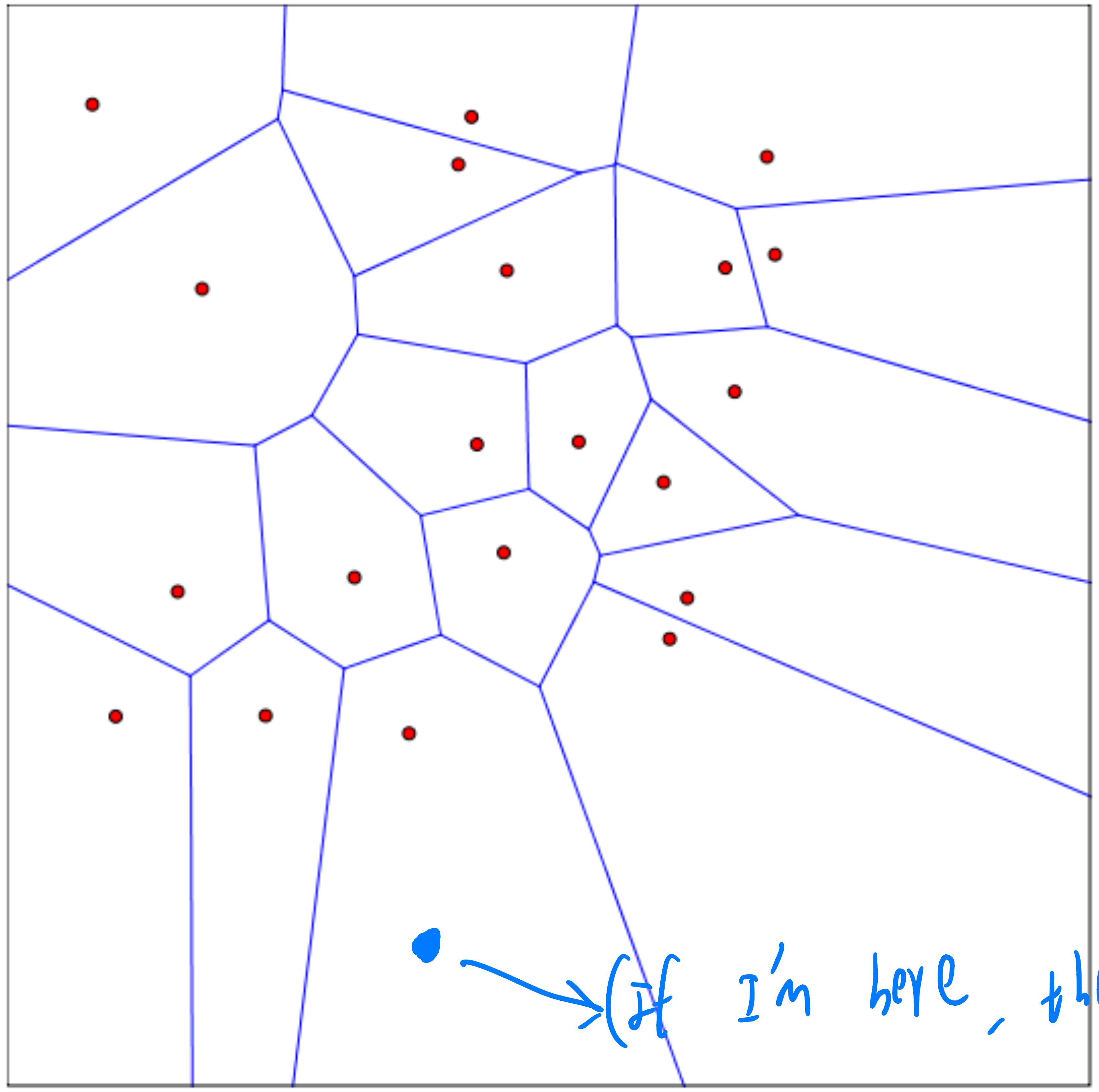
(a) classification by gender

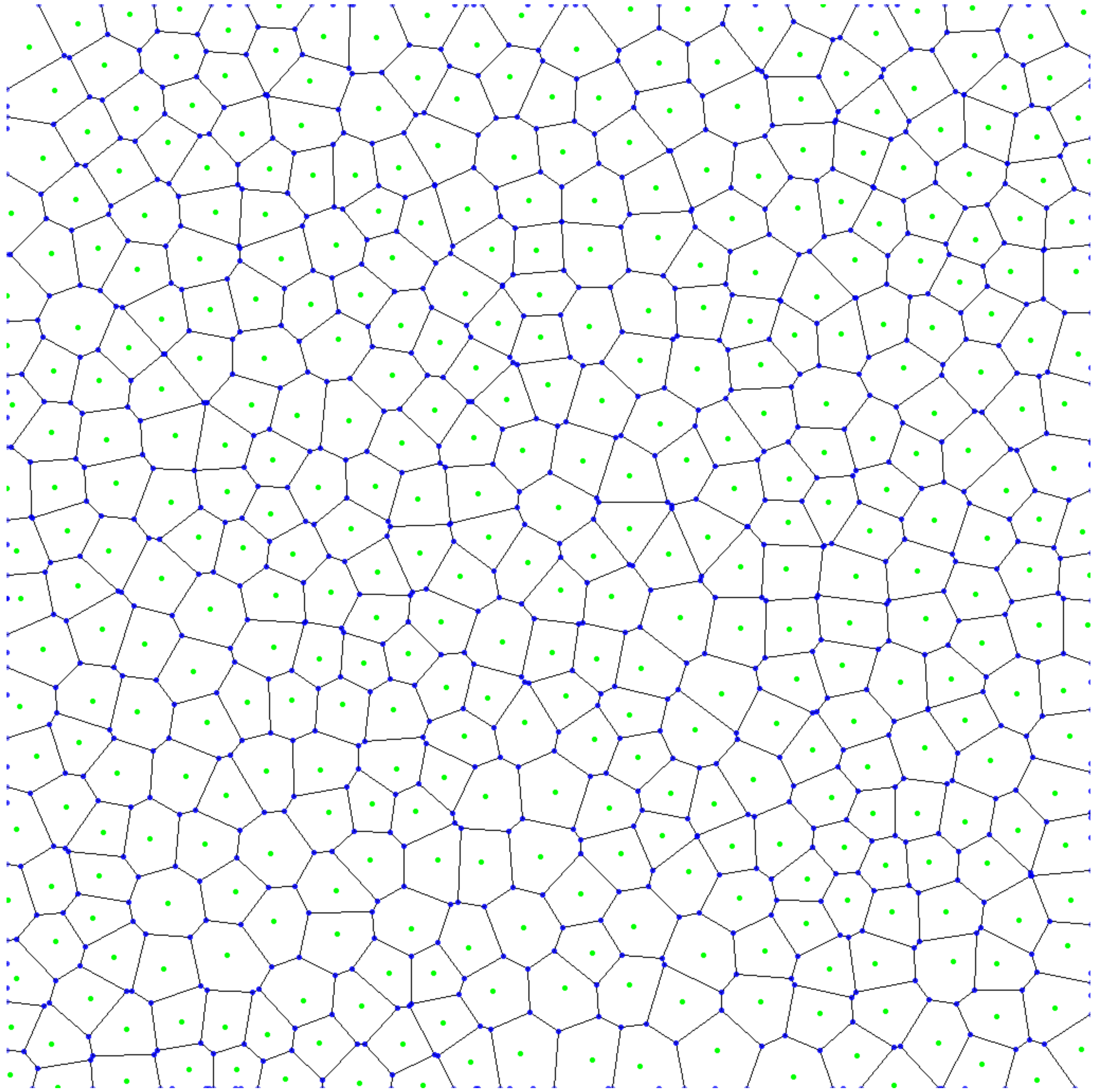
(b) classification by age

1NN: Decision Regions

Defined by "**Voronoi Diagram**"

- Each cell contains points closer to a particular training point





(if I'm here, the my class will be the center in this cell)

k-Nearest Neighbors: Distances

(Used when inputs are not in geometrical space but categorical / binary classes situation)

Discrete features: Hamming distance

For binary feature vectors.

$x^{(1)} = [0,1,1,0,0]^T$

$x^{(2)} = [1,0,1,0,0]^T$

$\rightarrow d_H(x^{(1)}, x^{(2)}) = 2$

(check each position is equal)

$$d_H(x^{(i)}, x^{(j)}) = \sum_{a=1}^{d} 1\{x_a^{(i)} \neq x_a^{(j)}\}$$

Continuous features:

- Euclidean distance:

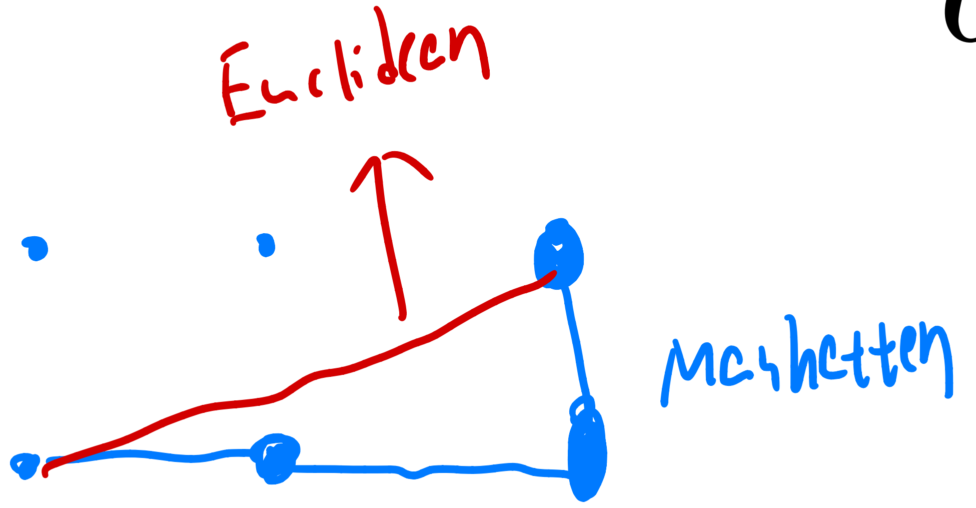
$$d(x^{(i)}, x^{(j)}) = \left(\sum_{a=1}^{d} (x_a^{(i)} - x_a^{(j)})^2 \right)^{\frac{1}{2}}$$

Two feature vectors

- L1 (Manhattan) dist.:

$$d(x^{(i)}, x^{(j)}) = \sum_{a=1}^{d} |x_a^{(i)} - x_a^{(j)}|$$

Absolute distance



k-Nearest Neighbors: Regression

Training/learning: given

$$\{(x^{(1)}, y^{(1)}), (x^{(2)}, y^{(2)}), \ldots, (x^{(m)}, y^{(m)})\}$$

Prediction: for x , find ***k*** most similar training points

Return

e.g. my 3-NN: $x^1 / x^2 / x^3$

Then my $\hat{y} = \frac{y^1 + y^2 + y^3}{3}$

$$\hat{y} = \frac{1}{k}\sum_{i=1}^{k} y^{(i)}$$

- I.e., among the ***k*** points, output mean label.

(Because in regression there is continuous values of y instead of a fixed set of discrete y)

More on **distance functions**…

- Be careful with **scale**
- Same feature but different units may change relative distance (fixing other features)
- Sometimes OK to normalize each feature dimension

Training set mean for dimension d

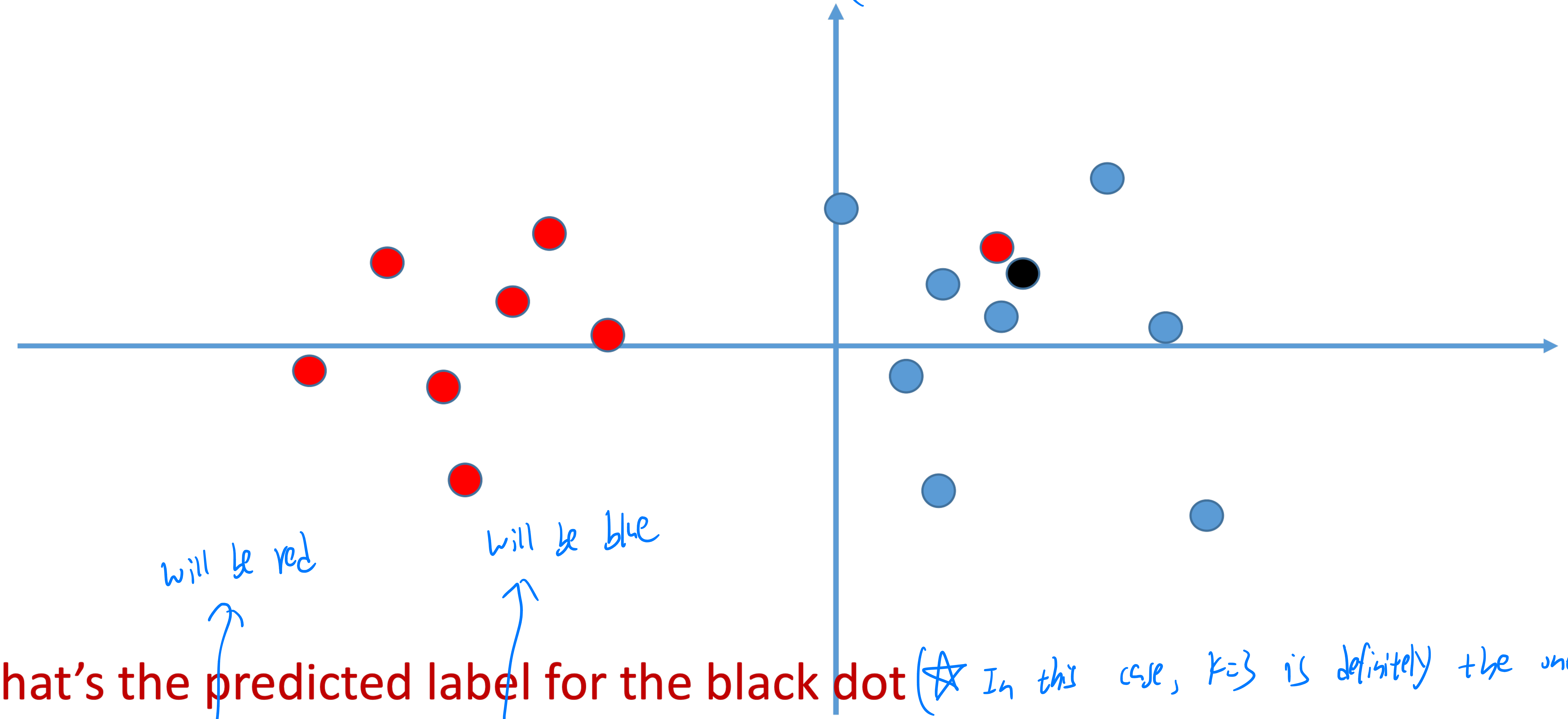
$$x'_{id} = \frac{x_{id} - \mu_d}{\sigma_d}, \forall i = 1 \ldots n, \forall d$$

Training set standard deviation for dimension d

- Other times not OK: e.g. dimension contains small random noise

Effect of k

k is important



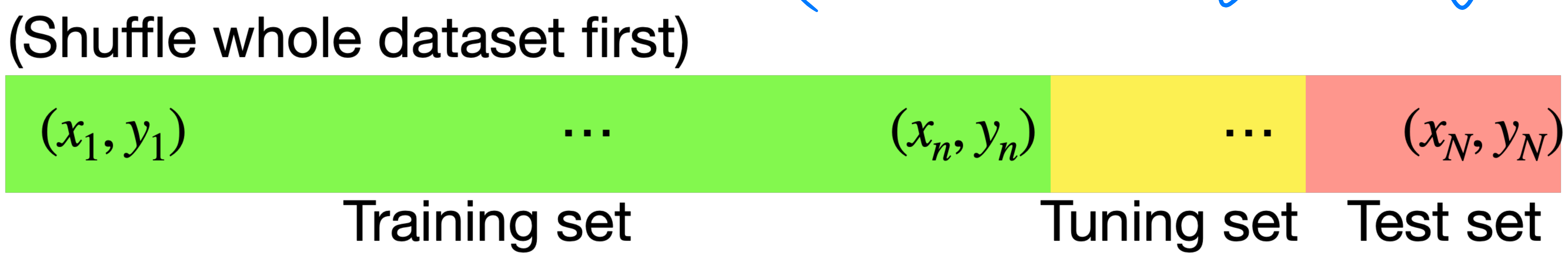
What's the predicted label for the black dot using 1 neighbor? 3 neighbors?

(☆ In this case, k=3 is definitely the one to go)

How to pick k, **the number of neighbors**

- Split data into training and **tuning sets**
- Classify tuning set with different k
- Pick k that produces least tuning-set error

Train / Test



★ procedure: k=1 train, k=1 tuning set

k=2 train, k=2 tuning set

⋮

Check what k is the most in tuning set

And finally test it using test set for fresh data

Quiz break

Q1-1: K-NN algorithms can be used for:

- A Only classification
- B Only regression
- C Both

Quiz break

Q1-1: K-NN algorithms can be used for:

- A Only classification
- B Only regression
- **C Both**

Quiz break

Q1-2: Which of the following distance measure do we use in case of categorical variables in k-NN?

- A Hamming distance
- B Euclidean distance
- C Manhattan distance

Quiz break

Q1-2: Which of the following distance measure do we use in case of categorical variables in k-NN?

- **A Hamming distance**
- B Euclidean distance
- C Manhattan distance

Quiz break

Q1-3: Consider binary classification in 2D where the intended label of a point x = (x1, x2) is positive if x1>x2 and negative otherwise. Let the training set be all points of the form x = [4a, 3b] where a,b are integers. Each training item has the correct label that follows the rule above. With a 1NN classifier (Euclidean distance), which ones of the following points are labeled positive? Multiple answers.

- [5.52, 2.41]
- [8.47, 5.84]
- [7,8.17]
- [6.7,8.88]

Quiz break

Q1-3: Consider binary classification in 2D where the intended label of a point x = (x1, x2) is positive if x1>x2 and negative otherwise. Let the training set be all points of the form x = [4a, 3b] where a,b are integers. Each training item has the correct label that follows the rule above. With a 1NN classifier (Euclidean distance), which ones of the following points are labeled positive? Multiple answers.

- **[5.52, 2.41]**
- **[8.47, 5.84]**
- [7,8.17]
- [6.7,8.88]

Nearest neighbors are
[4,3] => positive
[8,6] => positive
[8,9] => negative
[8,9] => negative
Individually.

Part II: Maximum Likelihood Estimation

Supervised Machine Learning

Non-parametric
(e.g., KNN)

vs.

Parametric

(Do not assume a specific form for the function)

(e.g. Linear Regression)

Supervised Machine Learning

Statistical modeling approach

Labeled training data (n examples)

$$(\mathbf{x}_1, y_1), (\mathbf{x}_2, y_2), \ldots, (\mathbf{x}_n, y_n)$$

drawn **independently** from
a fixed distribution
(also called the i.i.d. assumption)

Supervised Machine Learning

Statistical modeling approach

Labeled training data (n examples)	→	Learning algorithm	→	Classifier $\hat{f}$

$$(\mathbf{x}_1, y_1), (\mathbf{x}_2, y_2), \ldots, (\mathbf{x}_n, y_n)$$

drawn **independently** from
a fixed underlying distribution
(also called the i.i.d. assumption)

select $\hat{f}(\theta)$ from a pool of models $\mathcal{F}$
that **best describe the data observed**

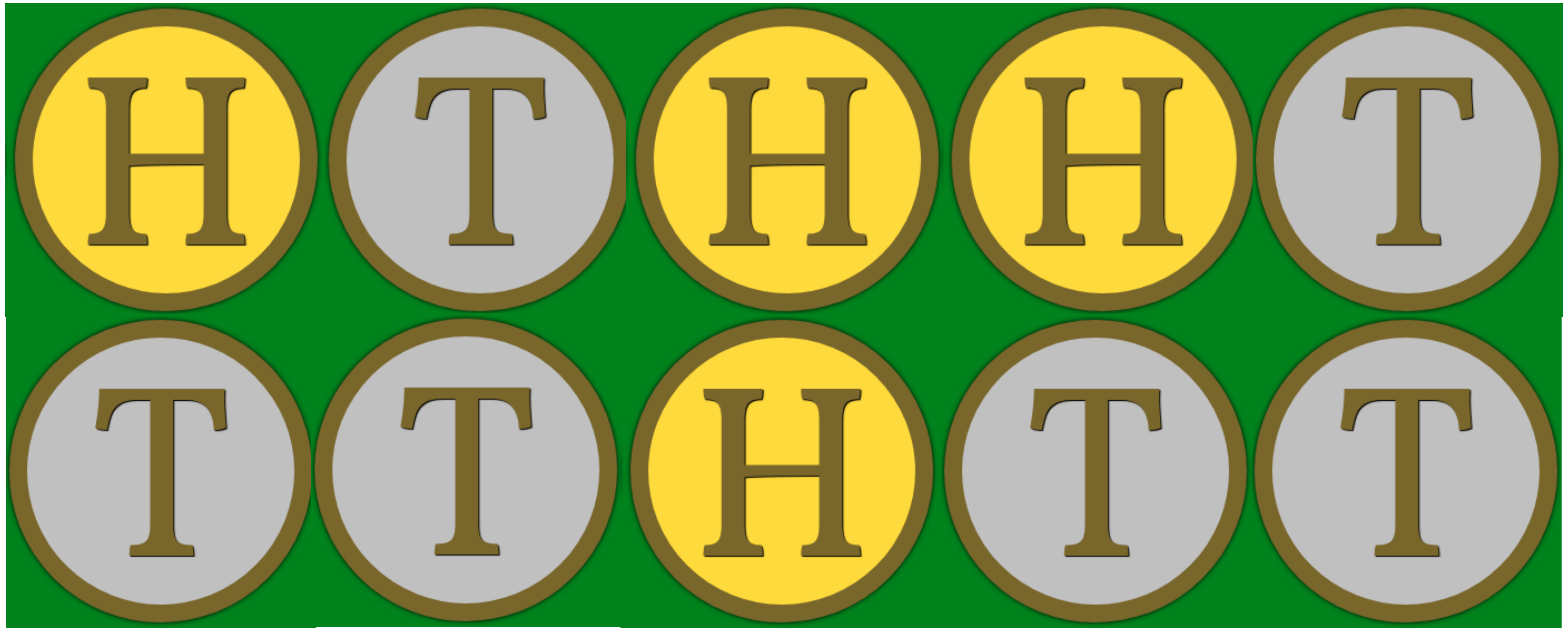
How to select $\hat{\boldsymbol{f}} \in \boldsymbol{\mathcal{F}}$?

- **Maximum likelihood (best fits the data)**
- Maximum a posteriori

  (best fits the data but incorporates prior assumptions)
- Optimization of 'loss' criterion (best discriminates the labels)

Maximum Likelihood Estimation: An Example

Flip a coin 10 times, how can you estimate $\theta = p(Head)$?



Intuitively, $\theta = 4/10 = 0.4$

How good is θ?

It depends on how likely it is to generate the observed data

$\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_n$

(Let's forget about label for a second)

Likelihood function $L(\theta) = \Pi_i p(\mathbf{x}_i | \theta)$

Under i.i.d assumption

Interpretation: How **probable** (or how likely) is the data given the probabilistic model p_θ?

☆ In MLE, θ is not coefficients like in linear regression, but a set of parameters that define a probability distribution. This gives confidence in our predictions

How good is θ?

It depends on how likely it is to generate the observed data $\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_n$

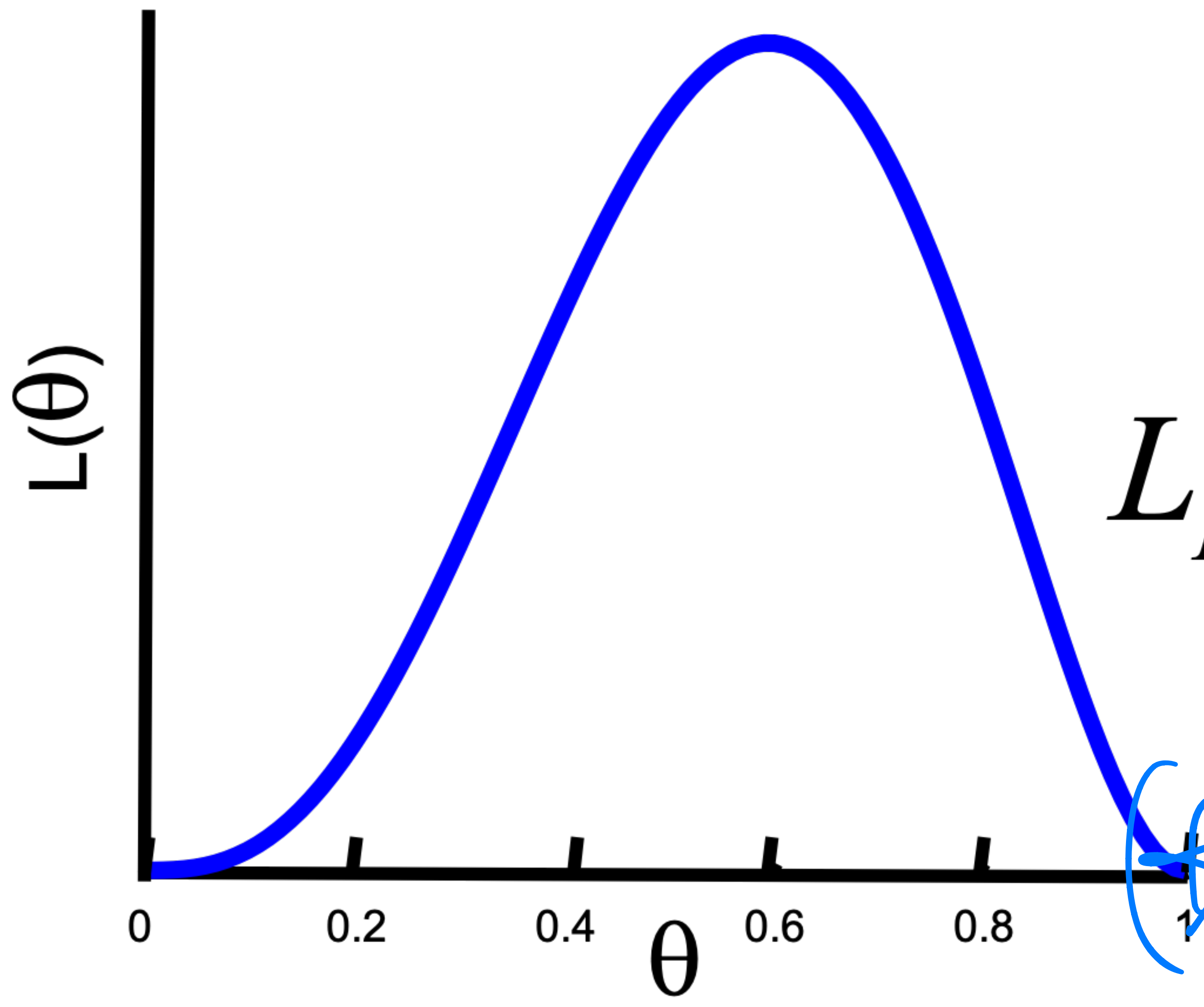
(Let's forget about label for a second)

Likelihood function $$L(\theta) = \Pi_i p(\mathbf{x}_i | \theta)$$

(In this case, θ is the probability of the coin being heads)

H,T, T, H, H

$$L_D(\theta) = \theta \cdot (1-\theta) \cdot (1-\theta) \cdot \theta \cdot \theta$$



(We want to maximize the Likelihood function)

Bernoulli distribution

Log-likelihood function

$$L_D(\theta) = \theta \cdot (1-\theta) \cdot (1-\theta) \cdot \theta \cdot \theta$$

$$= \theta^{N_H} \cdot (1-\theta)^{N_T}$$

Number of heads

Number of tails

Log-likelihood function

$$\ell(\theta) = \log L(\theta)$$

$$= N_H \log\theta + N_T \log(1-\theta)$$

Maximum Likelihood Estimation (MLE)

Find optimal θ^* to maximize the likelihood function (and log-likelihood)

$$\theta^* = \text{argmax} \underbrace{N_H \log\theta + N_T \log(1-\theta)}_{l(\theta)}$$

$$\frac{\partial l(\theta)}{\partial \theta} = \frac{N_H}{\theta} - \frac{N_T}{1-\theta} = 0 \Rightarrow \theta^* = \frac{N_H}{N_T + N_H}$$

(Derivative is 0 for maximum point)

which confirms your intuition!

No Test

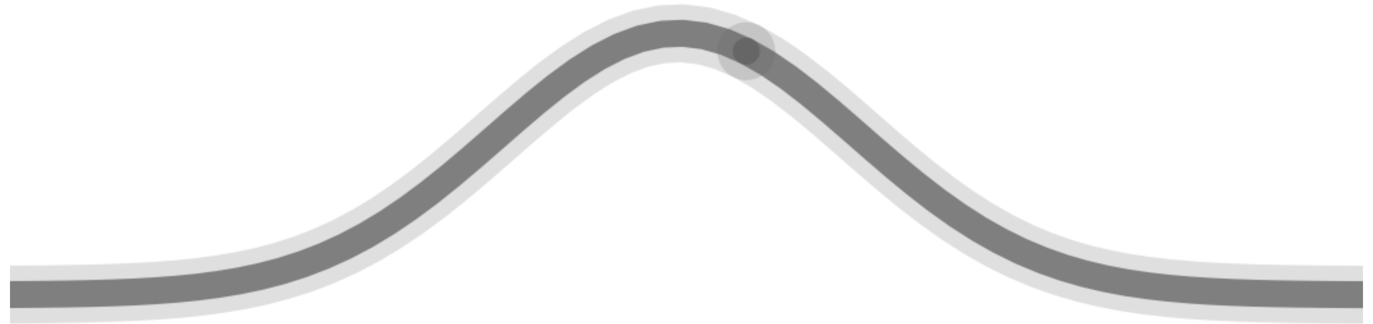
Maximum Likelihood Estimation: Gaussian Model

Fitting a model to heights of females

Observed some data (in inches): 60, 62, 53, 58,… $\in \mathbb{R}$

$$\{x_1, x_2, \ldots, x_n\}$$

Model class: Gaussian model



$$p(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

So, what's the MLE for the given data?

Estimating the parameters in a Gaussian

- **Mean**

$$\mu = \mathbf{E}[x] \quad \text{hence} \quad \hat{\mu} = \frac{1}{n}\sum_{i=1}^{n} x_i$$

- **Variance**

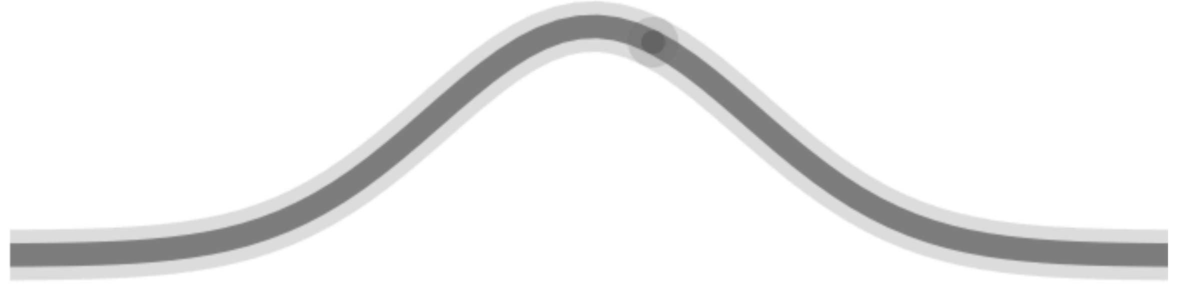
$$\sigma^2 = \mathbf{E}[(x-\mu)^2] \quad \text{hence} \quad \hat{\sigma}^2 = \frac{1}{n}\sum_{i=1}^{n} (x_i - \hat{\mu})^2$$

Why?

Maximum Likelihood Estimation: Gaussian Model

Observe some data (in inches): $x_1, x_2, \ldots, x_n \in \mathbb{R}$

Assume that the data is drawn from a Gaussian



$$L(\mu, \sigma^2 \,|\, X) = \prod_{i=1}^{n} p(x_i; \mu, \sigma^2) = \prod_{i=1}^{n} \frac{1}{2\pi\sigma^2} \exp\left(-\frac{(x_i - \mu)^2}{2\sigma^2} \right)$$

Fitting parameters is maximizing likelihood w.r.t μ, σ^2
(maximize likelihood that data was generated by model)

MLE

$$\arg\max_{\mu, \sigma^2} \prod_{i=1}^{n} p(x_i; \mu, \sigma^2)$$

Maximum Likelihood

- Estimate parameters by finding ones that explain the data

$$\underset{\mu,\sigma^2}{\operatorname{argmax}} \prod_{i=1}^{n} p(x_i;\mu,\sigma^2) = \underset{\mu,\sigma^2}{\operatorname{argmin}} - \log \prod_{i=1}^{n} p(x_i;\mu,\sigma^2)$$

- **Decompose likelihood**

$$\sum_{i=1}^{n} \frac{1}{2}\log(2\pi\sigma^2) + \frac{1}{2\sigma^2}(x_i-\mu)^2 = \frac{n}{2}\log(2\pi\sigma^2) + \frac{1}{2\sigma^2}\sum_{i=1}^{n}(x_i-\mu)^2$$

Minimized for $\mu = \frac{1}{n}\sum_{i=1}^{n} x_i$

Maximum Likelihood

- Estimating the variance

$$\frac{n}{2}\log(2\pi\sigma^2) + \frac{1}{2\sigma^2}\sum_{i=1}^{n}(x_i - \mu)^2$$

Maximum Likelihood

- Estimating the variance

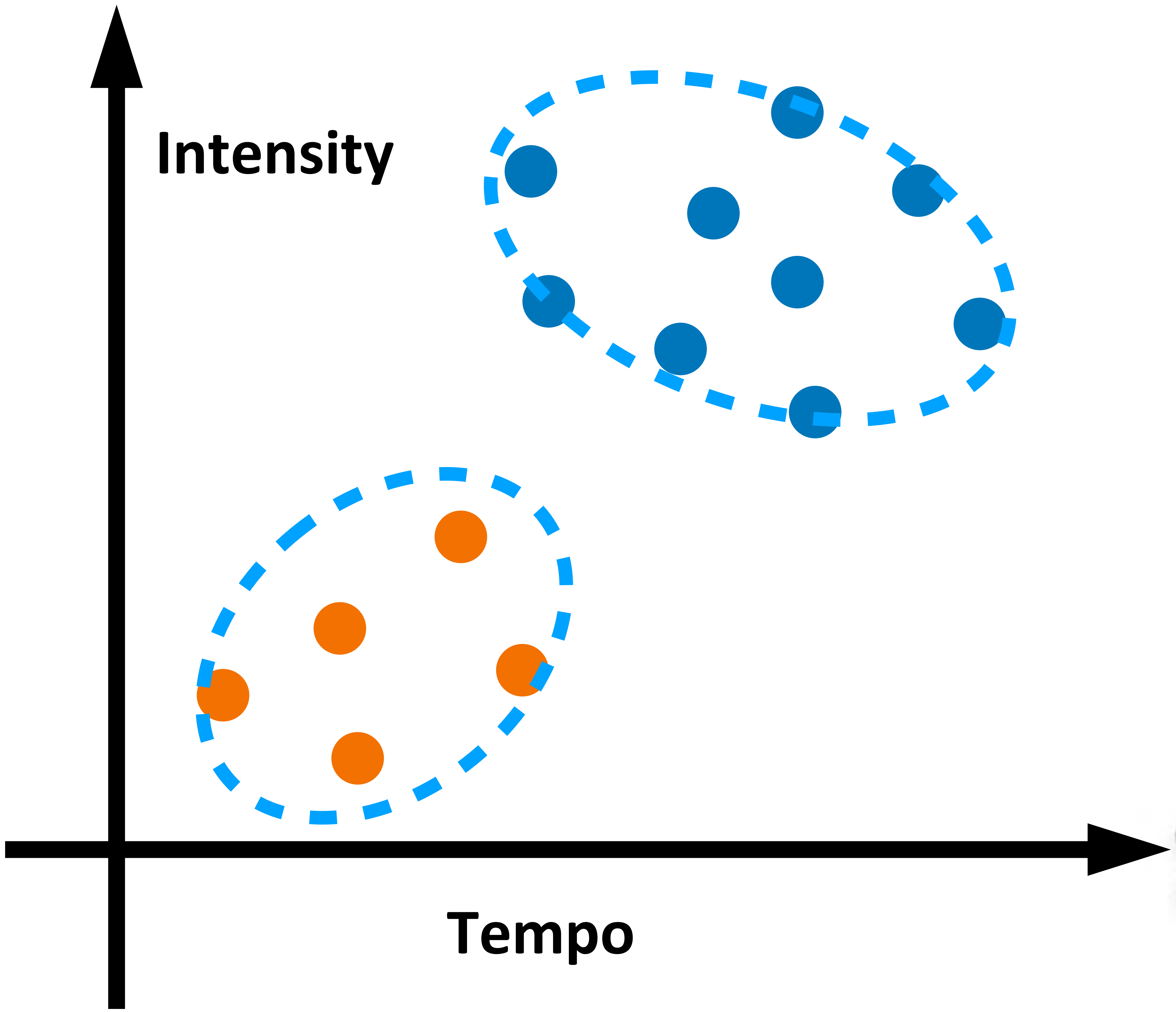
$$\frac{n}{2}\log(2\pi\sigma^2) + \frac{1}{2\sigma^2}\sum_{i=1}^{n}(x_i - \mu)^2$$

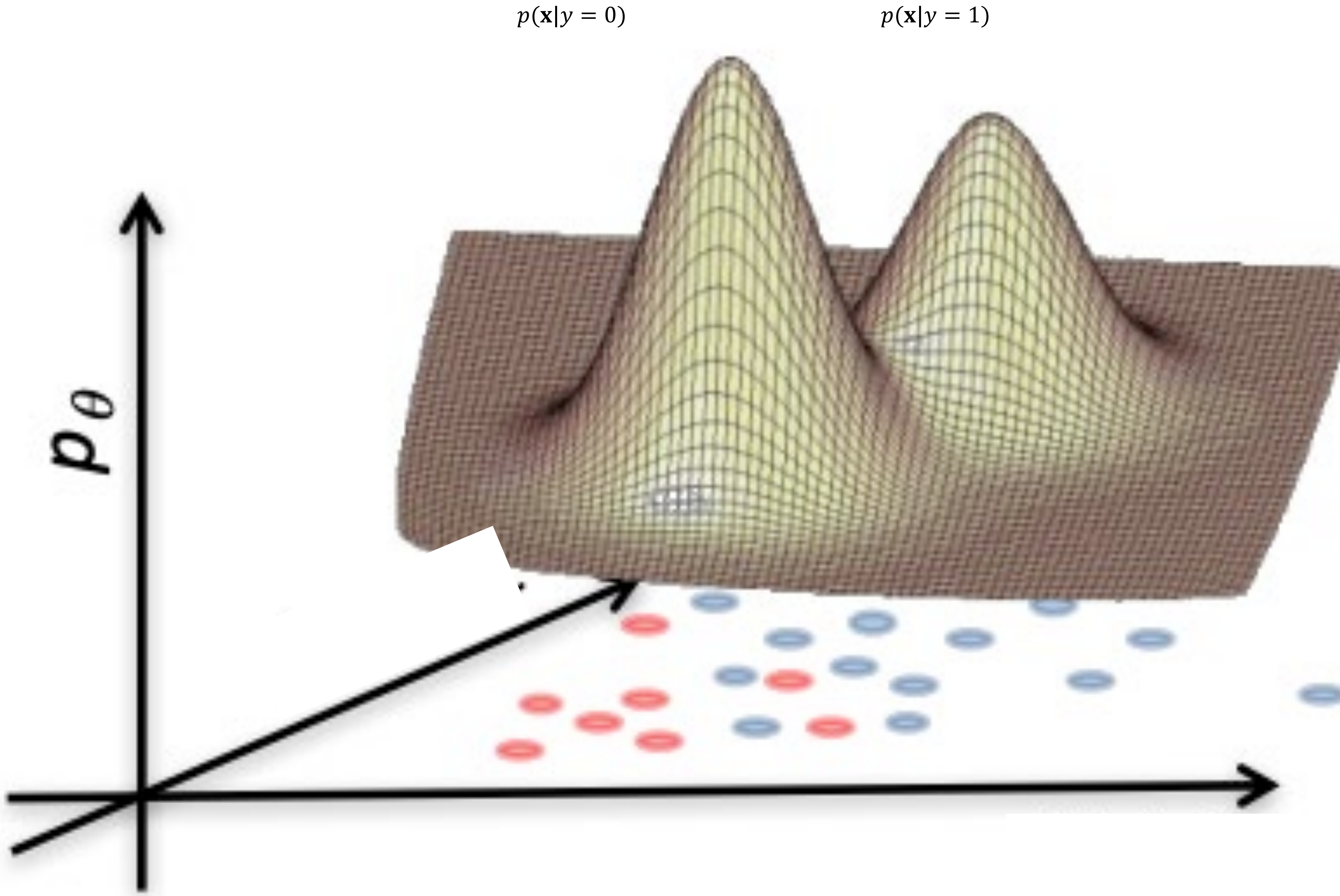
- Take derivatives with respect to it

$$\partial_{\sigma^2}[\,\cdot\,] = \frac{n}{2\sigma^2} - \frac{1}{2\sigma^4}\sum_{i=1}^{n}(x_i - \mu)^2 = 0$$

$$\implies \sigma^2 = \frac{1}{n}\sum_{i=1}^{n}(x_i - \mu)^2$$

Classification via MLE





Classification via MLE

$$\hat{y} = \hat{f}(\mathbf{x}) = \arg\max p(y \mid \mathbf{x})$$

(Posterior)

(Prediction)

Classification via MLE

$$\hat{y} = \hat{f}(\mathbf{x}) = \arg\max p(y \mid \mathbf{x})$$

(Prediction) (Posterior)

$$= \arg\max_{y} \frac{p(\mathbf{x} \mid y) \cdot p(y)}{p(\mathbf{x})}$$

(by Bayes' rule)

No y in it, irrelevant to our optimization on y

$$= \arg\max_{y} \underbrace{p(\mathbf{x} \mid y)}_{\text{likelihood}}\underbrace{p(y)}_{\text{prior}}$$

Using labelled training data, learn **class priors** and **class conditionals**

Quiz break

Q2-2: True or False
Maximum likelihood estimation is the same regardless of whether we maximize the likelihood or log-likelihood function.

- A True
- B False

Quiz break

Q2-2: True or False

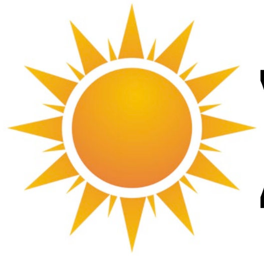
Maximum likelihood estimation is the same regardless of whether we maximize the likelihood or log-likelihood function.

- A True
- B False

Part III: Naïve Bayes

Example 1: Play outside or not?

- If weather is sunny, would you like to play outside?

Posterior probability p(Yes | ) vs. p(No | )

Example 1: Play outside or not?

- If weather is sunny, would you like to play outside?

Posterior probability p(Yes | ) vs. p(No | )

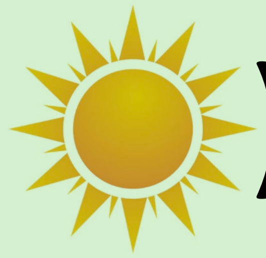
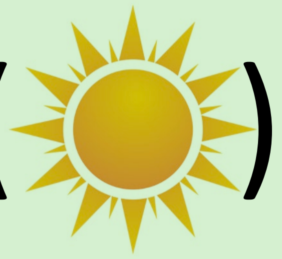
- Weather = {Sunny, Rainy, Overcast}
- Play = {Yes, No}
- Observed data {Weather, play on day *m*}, m={1,2,…,N}

Example 1: Play outside or not?

- If weather is sunny, would you like to play outside?

Posterior probability p(Yes | ) vs. p(No | )

- Weather = {Sunny, Rainy, Overcast}
- Play = {Yes, No}
- Observed data {Weather, play on day *m*}, m={1,2,…,N}

p(Play | ) = p( | Play) p(Play) / p()

Bayes rule

Example 1: Play outside or not?

- **Step 1**: Convert the data to a frequency table of Weather and Play

Weather	Play
Sunny	No
Overcast	Yes
Rainy	Yes
Sunny	Yes
Sunny	Yes
Overcast	Yes
Rainy	No
Rainy	No
Sunny	Yes
Rainy	Yes
Sunny	No
Overcast	Yes
Overcast	Yes
Rainy	No

Frequency Table		
Weather	No	Yes
Overcast		4
Rainy	3	2
Sunny	2	3
Grand Total	5	9

https://www.analyticsvidhya.com/blog/2017/09/naive-bayes-explained/

Example 1: Play outside or not?

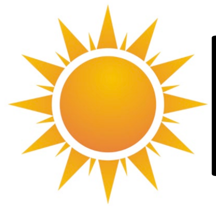
- **Step 1**: Convert the data to a frequency table of Weather and Play
- **Step 2**: Based on the frequency table, calculate **likelihoods** and **priors**

Weather	Play
Sunny	No
Overcast	Yes
Rainy	Yes
Sunny	Yes
Sunny	Yes
Overcast	Yes
Rainy	No
Rainy	No
Sunny	Yes
Rainy	Yes
Sunny	No
Overcast	Yes
Overcast	Yes
Rainy	No

Frequency Table		
Weather	No	Yes
Overcast		4
Rainy	3	2
Sunny	2	3
Grand Total	5	9

Likelihood table				
Weather	No	Yes		
Overcast		4	=4/14	0.29
Rainy	3	2	=5/14	0.36
Sunny	2	3	=5/14	0.36
All	5	9		
	=5/14	=9/14		
	0.36	0.64		

p(Play = Yes) = 0.64

p(| Yes) = 3/9 = 0.33

https://www.analyticsvidhya.com/blog/2017/09/naive-bayes-explained/

Example 1: Play outside or not?

- **Step 3**: Based on the likelihoods and priors, calculate posteriors

P(Yes|)

=P( |Yes)*P(Yes)/P()

?

P(No|)

=P( |No)*P(No)/P()

?

Example 1: Play outside or not?

- **Step 3**: Based on the likelihoods and priors, calculate posteriors

P(Yes| ☀)

=P(☀ |Yes)*P(Yes)/P(☀)

=0.33*0.64/0.36

=0.6

P(No|☀)

=P(☀ |No)*P(No)/P(☀)

=0.4*0.36/0.36

=0.4

P(Yes|☀) > P(No|☀) go outside and play!

Bayesian classification

$$\hat{y} = \arg\max p(y \mid \mathbf{x})$$ (Posterior)

(Prediction)

$$= \arg\max \frac{p(\mathbf{x} \mid y) \cdot p(y)}{p(\mathbf{x})}$$ (by Bayes' rule)

$$= \arg\max \; p(\mathbf{x} \mid y) p(y)$$

Bayesian classification

What if $\mathbf{x}$ has multiple attributes $\mathbf{x} = \{X_1, \ldots, X_k\}$

$$\hat{y} = \arg\max_y p(y \mid X_1, \ldots, X_k)$$

(Posterior)

(Prediction)

Bayesian classification

What if **x** has multiple attributes $\mathbf{x} = \{X_1, \ldots, X_k\}$

$$\boxed{\hat{y}} = \arg\max_y p(y \mid X_1, \ldots, X_k) \quad \text{(Posterior)}$$

(Prediction)

$$= \arg\max_y \frac{p(X_1, \ldots, X_k \mid y) \cdot p(y)}{\underbrace{p(X_1, \ldots, X_k)}_{\text{Independent of y}}} \quad \text{(by Bayes' rule)}$$

Bayesian classification

What if **x** has multiple attributes $\mathbf{x} = \{X_1, \ldots, X_k\}$

$$\hat{y} = \arg\max_y p(y \mid X_1, \ldots, X_k) \quad \text{(Posterior)}$$

(Prediction)

$$= \arg\max_y \frac{p(X_1, \ldots, X_k \mid y) \cdot p(y)}{p(X_1, \ldots, X_k)} \quad \text{(by Bayes' rule)}$$

$$= \arg\max_y \boxed{\underbrace{p(X_1, \ldots, X_k \mid y)}_{\text{Class conditional likelihood}} \; p(y)}$$

Class conditional likelihood

Class prior

Naïve Bayes Assumption

(Assume the independence among X)

Conditional independence of feature attributes

$$p(X_1, \ldots, X_k \mid y)p(y) = \Pi_{i=1}^{k} p(X_i \mid y) p(y)$$

Easier to estimate

(using MLE!)

Quiz break

Q3-1: Which of the following about Naive Bayes is incorrect?

- A Attributes can be nominal or numeric
- B Attributes are equally important
- C Attributes are statistically dependent of one another given the class value
- D Attributes are statistically independent of one another given the class value
- E All of above

Quiz break

Q3-1: Which of the following about Naive Bayes is incorrect?

- A Attributes can be nominal or numeric
- B Attributes are equally important
- **C Attributes are statistically dependent of one another given the class value**
- D Attributes are statistically independent of one another given the class value
- E All of above

Quiz break

Q3-2: Consider the following dataset showing the result whether a person has passed or failed the exam based on various factors. Suppose the factors are independent to each other.
We want to classify a new instance with
Confident=Yes, Studied=Yes, and Sick=No.

Confident	Studied	Sick	Result
Yes	No	No	Fail
Yes	No	Yes	Pass
No	Yes	Yes	Fail
No	Yes	No	Pass
Yes	Yes	Yes	Pass

- A Pass
- B Fail

Quiz break

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We want to classify a new instance with Confident=Yes, Studied=Yes, and Sick=No.

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Quiz break

We want to classify a new instance with Confident=Yes, Studied=Yes, and Sick=No.

- A Pass
- B Fail

Confident	Studied	Sick	Result
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Yes	No	Yes	Pass
No	Yes	Yes	Fail
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Yes	Yes	Yes	Pass

$$P(y = F|x_1 = Y, x_2 = Y, x_3 = N)$$
$$= \frac{1}{2} * \frac{1}{2} * \frac{1}{2} * \frac{2}{5} / P(x_1 = Y, x_2 = Y, x_3 = N)$$
$$\propto \frac{1}{4 * 5}$$

$$P(y = P|x_1 = Y, x_2 = Y, x_3 = N)$$
$$= \frac{P(x_1 = Y|Y = P)P(x_2 = Y|Y = P)P(x_3 = N|Y = P)P(y = P)}{P(x_1 = Y, x_2 = Y, x_3 = N)}$$
$$= \frac{2}{3} * \frac{2}{3} * \frac{1}{3} * \frac{3}{5} / P(x_1 = Y, x_2 = Y, x_3 = N)$$
$$\propto \frac{4}{9*5}$$

Larger!

What we've learned today…

- K-Nearest Neighbors
- Maximum likelihood estimation
  - Bernoulli model
  - Gaussian model
- Naive Bayes
  - Conditional independence assumption

Thanks!